

What are Tasks in Snowflake?

A **Task** in Snowflake is used to **automate SQL operations**.

Tasks allow users to **schedule and execute SQL statements automatically** without manual intervention.

Tasks are commonly used for:

- Automating data pipelines
- Running scheduled SQL queries
- Processing incremental data
- Running ETL / ELT jobs

Tasks are often used together with **Streams** to process **new or changed data automatically**.

Key Concept of Tasks

A **Task** **executes a SQL statement based on a defined schedule or dependency**.

Tasks can run:

- At a **specific time interval**
- After **another task finishes**

This helps build **automated workflows and data pipelines** in Snowflake.

Why Tasks are Used

Tasks help automate **data engineering workflows**.

Instead of manually running queries, tasks can:

- Load data automatically
- Transform data regularly
- Process CDC data from streams
- Run scheduled reports

This makes Snowflake capable of handling **fully automated ELT pipelines**.

Types of Task Execution

Snowflake tasks can run in **two ways**.

1 Scheduled Tasks

These tasks run based on a **defined schedule using CRON expressions or time intervals**.

Example:

Run every hour.

```
CREATE TASK hourly_task
WAREHOUSE = compute_wh
SCHEDULE = '1 HOUR'
AS
INSERT INTO target_table
SELECT * FROM source_table;
```

This task runs **every 1 hour automatically**.

2 Triggered Tasks (Task Dependencies)

Tasks can also run **after another task completes**.

This creates a **task chain or workflow**.

Example:

Task1 → Load data

Task2 → Transform data

Task3 → Aggregate data

This builds a **pipeline inside Snowflake**.

Example Task Creation

Example:

```
CREATE TASK load_sales_task
WAREHOUSE = compute_wh
SCHEDULE = 'USING CRON 0 * * * * UTC'
AS
INSERT INTO sales_target
SELECT * FROM sales_stage;
```

Explanation:

- Task runs **every hour**
 - Uses **compute warehouse**
 - Executes SQL automatically
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Starting a Task

After creating a task, it must be **resumed to start execution**.

```
ALTER TASK load_sales_task RESUME;
```

If the task is suspended, it will not run.

Suspending a Task

To stop a task:

```
ALTER TASK load_sales_task SUSPEND;
```

This pauses the scheduled execution.

Task Dependencies (Task Trees)

Snowflake supports **task chaining**.

Example workflow:

Task1 → Load raw data

Task2 → Transform data

Task3 → Load into fact table

Example:

```
CREATE TASK transform_task  
AFTER load_task  
AS  
INSERT INTO clean_table  
SELECT * FROM raw_table;
```

This task runs **only after load_task completes successfully**.

Tasks with Streams (Most Common Real-Time Use Case)

The **most common real-world use case** is:

Streams + Tasks

Example workflow:

- 1 New data inserted into table
- 2 Stream captures the change
- 3 Task processes the stream automatically
- 4 Data loaded into another table

Example:

```
CREATE TASK process_stream_task
WAREHOUSE = compute_wh
SCHEDULE = '5 MINUTE'
WHEN
SYSTEM$STREAM_HAS_DATA('sales_stream')
AS
INSERT INTO sales_processed
SELECT * FROM sales_stream;
```

Explanation:

- Task runs every **5 minutes**
- Checks if stream has new data
- Processes only changed records

This creates a **real-time incremental pipeline**.

Key Characteristics of Tasks

- Tasks automate SQL execution
 - Tasks require a **virtual warehouse** to run
 - Tasks can run on **schedules or dependencies**
 - Tasks support **task chaining**
 - Tasks are often used with **streams**
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Monitoring Tasks

Snowflake allows monitoring task execution.

Commands:

```
SHOW TASKS;
```

View task history:

```
SELECT *
FROM TABLE(INFORMATION_SCHEMA.TASK_HISTORY());
```

This shows:

- Execution status
 - Start time
 - End time
 - Errors
-

Task Tree Visualization

Snowflake allows viewing **task dependency graphs**.

Command:

```
SELECT *  
FROM TABLE(INFORMATION_SCHEMA.TASK_DEPENDENTS('task_name'));
```

This helps understand **workflow pipelines**.

Important Limitations

- Tasks require a **warehouse to run**
 - Tasks must be **resumed manually after creation**
 - Failed tasks must be **monitored and fixed**
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Typical Interview Questions

Question

Which Snowflake object is used to automate SQL operations?

Answer:

Tasks

Question

Do tasks require a warehouse to execute?

Answer:

Yes

Tasks use **virtual warehouses for compute**.

Question

Can tasks run automatically on a schedule?

Answer:

Yes

Using **CRON expressions or time intervals**.

Question

Can tasks depend on other tasks?

Answer:

Yes

Snowflake supports **task chaining**.

Question

What feature is commonly used with Tasks for incremental data pipelines?

Answer:

Streams

Quick Summary

Tasks in Snowflake allow **automation of SQL queries and data workflows**.

They enable **scheduled execution, workflow orchestration, and real-time data processing** when combined with Streams.

Tasks are a key component in building **fully automated ELT pipelines in Snowflake**.